

Faculty of Science, Engineering and Built Environment

SEJ302 Control Systems Engineering

Deakin University Unit Guide

Trimester 2, 2025

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Welcome

This project-oriented design-based learning (PODBL) unit has been developed with significant industry input. The learning and assessment activities in the unit require students to explore and apply control systems theory on physical systems to develop practical skills and deeper theoretical knowledge, while also working as part of a team to deliver a well justified and validated control systems design project for a customer.
Students will actively participate in labs throughout the trimester that require them to work with electromechanical and fluid control systems. In these labs students will explore, apply, and communicate control systems concepts and skills such as theoretical and mathematical system modelling, Laplace and state space system modelling, system linearisation, PID control design and parameter tuning, stability analysis, and practical applications of instrumentation, monitoring, and data acquisition devices.
Project and individual work will require students to design and evaluate controllers for a variety of interfacing systems to solve a real-world problem. This will allow students to explore and create designed control system solutions for various dynamic systems by applying fundamentals of control systems theory, such as transfer function derivation, steady state error analysis, root locus and Bode methods for system modification and analysis, PID's, and control system evaluation using simulation tools such as MATLAB/SIMULINK. Students will work in small teams to plan, manage, and report project deliverables to meet customer expectations related to cost, functionality, hardware selection, resourcing, and timing.
This Unit Guide provides you with the key information about this unit. Please read it carefully and refer to it frequently throughout the study period. Your unit site also provides information about your rights and responsibilities. We will assume you have read this before the unit commences, and we expect you to refer to it throughout the study period.

To be successful in this unit, you must:

- read all materials in preparation for your learning activities and follow up each with further study and research on the topic
- start your assessment tasks well ahead of the due date
- read or listen to all feedback carefully and use it in your future work
- attend and engage in all educator facilitated (scheduled) learning activities and other learning experiences as part of the unit design

Who is the unit team?

Unit chair: leads the teaching team and is responsible for overall delivery of this unit

Name: Rob McHenry
Campus: Geelong Waurn Ponds
Email: r.mchenry@deakin.edu.au
Phone: +61 3 522 73845
Other members of the team and how to contact them

Seminar and Studio Teaching Staff

Name: Dr Michael Norton
Campus: Geelong Waurn Ponds
Email: michael.norton@deakin.edu.au
Phone: +61 3 524 79476

Name: Dr. Van Thanh Huynh
Campus: Geelong Waurn Ponds
Email: v.huynh@deakin.edu.au

Laboratory Demonstrator

Name: Sheikh Tanzim Meraj

Campus: Geelong Waurn Ponds
Email: s.meraj@deakin.edu.au
Administrative queries

- check-out the 'SEBE Student Hub' section on your unit site
- contact your Unit Chair or Campus Leader
- drop in or contact Student Central to speak with a Student Adviser

For additional support information, please see the Rights and Responsibilities section under 'Content' in your unit site.
About this unit

Unit development in response to student feedback

Every trimester, we ask students to tell us, through eVALUate, what helped and hindered their learning in each Unit. You are strongly encouraged to provide constructive feedback for this Unit when eVALUate opens (you will be emailed a link).
In previous versions of this unit, students have told us that these aspects of the unit have helped them to achieve the learning outcomes:

Practical application of control systems in labs helped visual concepts learned in class and projects:

- Physical systems like the servo and coupled tanks helped students visualise their work.
- Labs provided a tangible link between theory and application, enhancing understanding.

The project's industry context helped develop useful project management and technical skills for employment:

- Industry-informed projects and guest lectures in seminars provided real-world context.
- Project management elements were well-integrated, preparing students for post-graduation roles.

Engaging with the learning resources, seminars, and studios was helpful:

- Weekly technical seminars and online videos were effective for understanding control systems theory.
- Learning resources and seminars reinforced theoretical concepts and allowed for interactive learning.

The following aspects of the unit have been introduced, enhanced or retained in response to feedback from students who have undertaken this unit in previous trimesters:

- Assessment has aligned learning been redesigned in the following ways in response to feedback from students:
    - Laboratory assessment task expectations have been reduced slightly particularly for the first submission to allow students more time to become familiar with control systems theory and working with lab equipment, and to reduce perceptions of high unit study load.
    - Project control system design requirements and laboratory system learning alignment has been improved (i.e. similar systems will be controlled in the project to the ones students work on in labs).
    - Final Team and Individual Project deliverables scope has been reduced slightly from 2024 due to changes in week 4 assessment focus (see below) and to reduce high workload at end of trimester.
    - There has been a reduction in focus on reflective practice activities, however, reflective practice and providing quality feedback to peers are still core elements of assessment.
    - Project Team formation has been pushed back to weeks 5 and 6 to allow students more time to work on building their capacity to model plant systems in preparation for designing controllers (big challenge for students in 2024), and to allow time to better understand project requirements.
    - Students will now create and evaluate their own system models from scratch in weeks 1 - 4 and be assessed on their system modelling by both staff and peers.
    - Additional learning support for system modelling has been introduced in weeks 1 - 3 in support of above changes.

If you have any concerns about the Unit during the trimester, please contact the unit teaching team - preferably early in the

trimester - so we can discuss your concerns, and make adjustments, if appropriate.

Learning Outcomes

Each Unit in your course is a building block towards Deakin's Graduate Learning Outcomes - not all Units develop and assess every Graduate Learning Outcome (GLO).

ULO	These are the Unit Learning Outcomes (ULOs) for this unit. At the completion of this unit, successful students can:	Alignment to Deakin Graduate Learning Outcomes (GLOs)
ULO1	Apply control systems theory on physical systems to communicate and demonstrate practical and problem-solving skills individually and in collaboration with others.	GLO1: Discipline-specific knowledge and capabilities GLO2: Communication GLO5: Problem solving GLO7: Teamwork
ULO2	Apply the fundamentals of mathematical modelling, characterisation, and evaluation of control systems using simulation tools to design and evaluate controllers for a variety of dynamic systems.	GLO1: Discipline-specific knowledge and capabilities GLO4: Critical thinking
ULO3	Work as part of a team to deliver a well justified and validated control systems design project that meets customer expectations related to cost, functionality, hardware selection, resourcing, and timing.	GLO1: Discipline-specific knowledge and capabilities GLO2: Communication GLO4: Critical thinking GLO7: Teamwork
ULO4	Manage personal work and deliverables by planning, performing, and reflecting upon personal contributions to team outcomes including professional judgement and ethical decision making.	GLO6: Self-management GLO8: Global citizenship Self-management

Assessing your achievement of the unit learning outcomes

Hurdle requirements

To be eligible to obtain a pass in this unit, students must achieve an overall mark of at least 50% in the Individual portion of project portfolio AT2: Team control systems design project portfolio as well as AT3: Control systems lab portfolio.

Brief summary of the hurdle requirements	Rationale
Students must achieve a mark of at least 50% in the individual portion of the Assessment 2: Team control systems design project portfolio	This is to ensure individuals are contributing to team outcomes in project work by delivering their technical and project responsibilities (ULO2), while working as a team to deliver a validated, cohesive, and well justified project outcome (ULO3). Students will also need to evidence how they can plan, perform, and reflect upon personal contributions to deliver team outcomes by evidencing professional judgement and ethical decision making (ULO4).
Students must achieve a mark of at least 50% in Assessment 3: Control systems lab portfolio	This unit requires a pass in the control systems laboratory portfolio to ensure that each student can apply control systems theory on practical applications that require them to communicate complex ideas and collaborate with others (ULO1), while managing their own personal work, decision making and contributions to those collaborative outcomes (ULO4). These are critical capabilities that graduates are expected to be able to demonstrate and evidence by employers consulted in development of this unit.

Summative assessment (tasks that will be graded or marked)

NOTE: It is *your responsibility* to keep a backup copy of every assignment and the materials used to develop/complete it where possible (e.g. written/digital reports, essays, videos, images). In the unusual event that one of your submissions becomes corrupted, is incorrectly submitted or otherwise lost, you may be asked to submit the backup copy. Any work you submit may be checked by electronic or other means for the purposes of detecting breaches of academic integrity such as collusion, plagiarism and contract cheating. You must understand your responsibility to act with honesty and integrity in your studies as Deakin takes all breaches very seriously. Make sure you read Your rights and responsibilities as a

student in this unit to find out more about academic integrity.

Deakin has a universal assessment submission time of 8 pm AEDT/AEST. A late penalty will apply to assessments submitted after 11.59 pm AEDT/AEST.

Summative assessment task 1

Title	**Control systems theory tests**
Brief description of assessment task	Regular (approximately fortnightly) submission of proposed solution to given analytical problems that test ability to apply theoretical knowledge (multiple attempts allowed)
Detail of student output	Approximately every two weeks, students will be provided with one or two analytical problems that require them to draw upon theoretical learning obtained through engagement with weekly seminars and online learning content. Theoretical knowledge gained through these activities will directly support student learning on concepts relevant to both Assessment tasks 1 and 2. Students will be able to use these tasks to gauge their ability to apply theoretical concepts accurately in a more well-defined situation to develop confidence in their other work on assessments 2 and 3. Each problem will be assessed on a pass / fail basis. Students are encouraged to ask questions from technical experts in their weekly seminar to help develop understanding of, and ability to apply the theoretical concepts being developed in these tests.
Grading and weighting (% total mark for unit)	10% (2%-4% per submission)
This task assesses your achievement of these Unit Learning Outcome(s)	ULO2: These submissions will test students ability to accurately apply the fundamentals of mathematical modelling, characterisation, and evaluation of control systems theory in support of developing their ability to use simulation tools, work on physical systems, and deliver project outcomes.
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1: Discipline-specific knowledge and capabilities through application of core theoretical concepts to solve a given problem. GLO4: Critical thinking developed through your ability to identify and apply the correct theoretical concepts to each proposed problem.
How and when you will receive feedback on your work	Students will receive feedback on their assessment via the unit site, normally within 10 working days of the submission due date.
When and how to submit your work	Due 8pm (AEST), Friday of weeks 3, 5, 8 and 10. Online submission via the unit site.

Summative assessment task 2

Title	**Team control systems design project**
Brief description of assessment task	Students will work individually and in teams to explore various control system design approaches for several different interfacing sub-systems to design a series of justified control system solutions that are required to meet a customer design brief on a real-world representative system. Each student will be responsible for leading the controller design of at least one sub-system each and will need to work as a team to deliver a justified, coherent, and validated engineering solution that considers all technical and social constraints.

Detail of student output	Students are expected to deliver a portfolio of activities both individually and as a team. This includes but may not be limited to: • Sub system plant simulation model development and validation. • Giving and receiving feedback to peers on work quality. • Formal documentation of team responsibilities and project management deliverables. • Control system design and justification for a minimum number of interfacing subsystems to meet technical, financial, and resource needs of the customer. • Design validation and verification evidence and reporting, including sign-off documentation and team approvals. • Project milestone management check-ins conducted in studios and supported with written evidence. • Reflective artifacts and documentation of lessons learned both individually and as a team.
Grading and weighting (% total mark for unit)	40% (20% team, 20% individual)
This task assesses your achievement of these Unit Learning Outcome(s)	ULO2: Each student will apply the fundamentals of mathematical modelling, characterisation, and evaluation of control systems using simulation tools to design and evaluate controllers for a variety of dynamic systems to deliver project outcomes. ULO3: Each student is required to work as part of a team to deliver their well justified and validated control systems designs within the project to meet customer expectations related to cost, functionality, hardware selection, resourcing, and timing. ULO4: Each student will need to evidence their ability to manage personal work and deliverables, plan, and deliver for their team by reflecting upon personal contributions to team outcomes, professional judgements, and ethical decision making.
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1: Discipline-specific knowledge and capabilities through appropriate application of control system theory and tools to solve engineering problems. GLO2: Communication evidenced by written and verbal submissions, as well as reflections on how effectively you and others have communicated. GLO4: Critical thinking will be evidenced by your ability to justify your decisions and validate your proposed control system solutions. GLO6: Self-management through evidence of planning, and reflection on your impact on team outcomes and peer feedback. GLO7: Teamwork through the coherence of your individual and team submissions, and evidence of teamwork activities submitted throughout the trimester. GLO8: Global citizenship by evidencing your ability to apply your social responsibilities as an engineer within your design justification that accounts for requirements of your stakeholders and wider society.
How and when you will receive feedback on your work	Students will receive feedback on this assessment task via the unit site within two weeks of submission.
When and how to submit your work	There will be regular submissions as indicated by project milestones throughout the trimester in weeks 4, 9 and 12. Submissions will be both via the unit site and within studio times. Details to be shared via the unit site.

Summative assessment task 3

Title	**Control systems lab portfolio**

Brief description of assessment task	Students are expected to engage in and complete two (2) lab sequences of learning and assessment across the trimester. Students are to complete the two sequences using different lab hardware from the selection of electromechanical and fluid control systems available. Digital twin versions of lab systems are available online for all students to access for unlimited use during the trimester. Online students will be expected to complete Lab sequence A activities during Week 6 (on-campus), and Lab sequence B using a digital twin system.
Detail of student output	Students are expected to engage in and complete the following two (2) lab sequences. The order in which students complete these sequences will vary based on limited physical system availability. **Lab sequence A:** • Students will have the option to choose either an electromechanical or fluid control system to explore and learn control systems concepts and get familiar with engaging in laboratory learning. • Students will spend four (4) lab sessions (8 hours equivalent) working through a series of activities specific to their selected equipment to develop an understanding of how the system is modelled, how the real system behaves, and how control systems principles can be applied to enable the system to perform increasingly complex functions. • Towards the end of the sequence, students will engage in a 5 - 10 minute verbal assessment of their knowledge and demonstration of their skills with the lab demonstrator. • Students will submit a laboratory report that evidences quality of learning and progress, including explanations of key concepts, explanations of how concepts translate and apply to real systems, documentation of methods and outcomes, evidence of raw notes, and reflection upon learning from their lab demonstrator discussion.
	Lab sequence B: As per sequence A with the following changes and additions: • Students must choose a different or more complex system to explore in sequence B compared to sequence A. • Students will spend three (3) lab sessions (6 hours equivalent) working through dedicated lab activities on their selected hardware for sequence B. • Students will spend one (1) lab session working with a partner to create, develop, and test a work instruction for their selected hardware. • Students will undertake a lab demonstrator discussion and submit a lab report as per sequence A. • In addition, students will submit their final work instruction and a short reflection (500 words max) explaining how their collaboration with a partner informed their approach to creating a work instruction.
Grading and weighting (% total mark for unit)	50%
This task assesses your achievement of these Unit Learning Outcome(s)	ULO1: Students will engage in the application of control systems theory on physical systems to gain a deeper understanding and communicate and demonstrate practical and problem-solving skills individually and in collaboration with others. ULO2: Students will apply the fundamentals of control systems modelling, characterisation and evaluation in labs on a variety of dynamic systems. ULO4: Students will need to manage their own work progress, make judgements, and decide on their approach to managing personal contributions in labs that will impact the performance of others.

This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1: Discipline-specific knowledge and capabilities by documenting evidence of engagement and exploration of theoretical concepts in practical applications. GLO2: Communication assessed both written and verbally, through documentation of work, creating of work instructions, and providing to and responding to feedback. GLO4: Critical Thinking by demonstrating your ability to think your way through practical problems, and inform next steps while applying appropriate skills and theoretical knowledge. GLO5: Problem solving by demonstrating your ability to identify what problem needs to solved, and evidencing how you approached solving it. GLO6: Self-management by maintaining steady progress throughout the labs to complete required tasks, and be aware of your abilities and limitations when working with others. GLO7: Teamwork through both written and social collaboration tasks that are required in the fourth session of each lab sequence. GLO8: Global Citizenship by explaining how the skills and tools you have learned can be used in wider social and technical contexts that could affect society or the environment.
How and when you will receive feedback on your work	Students will receive feedback via the unit site within 2 weeks of submitting the portfolio for each lab sequence.
When and how to submit your work	There will be two (2) lab sequence portfolio submission dates. • **Lab sequence A** submission date: 8pm Sunday of Week 7, 31 August 2025. • **Lab sequence B** submission date: 8pm Monday of Week 12, 29 September 2025. On-campus students will attend 2-hour lab sessions in weeks 2 - 5, and 7 - 10, and will have access to online digital twin systems to explore in their own time. Online students will attend an online weekly drop-in session (combined with online studio) from weeks 2 - 5, and 7 - 10 to prepare for week 6 on-campus labs and support their completion of Lab sequence B using an online digital twin system. Online students are expected to attend a 2-day intensive activity in week 6 to complete Lab sequence A using a physical system. Further details will be provided on the unit site.

Your learning experiences in this unit

Educator-facilitated (scheduled) learning activities - on-campus unit enrolment

1 x 1 hour online content per week (learning modules), 2 x 1 hour seminar per week, 1 x 2 hour Practical experience (studio) per week, 1 x 2 hour practical experience (laboratory) per week in weeks 2 – 5, and 7 – 10.

Educator-facilitated (scheduled) learning activities - online unit enrolment

1 x 1 hour online content per week (learning modules); 2 x 1 hour online seminar per week; 2 x 1 hour online Practical experience (studio) per week and 1 x 2 day intensive lab activity in week 6.

Typical study commitment

Students will on average spend 300 hours over the teaching period undertaking the teaching, learning and assessment activities for this unit.

This will include educator guided online learning activities within the unit site.

This two credit point unit is informed by observations of industry practice and setup to mimic the role of a graduate control systems engineer as much as possible. As such you are expected to develop your individual expertise in control systems engineering, while delivering core project outcomes for your employer, and improving your organisations hardware capabilities (through practical studio's). This means you are expected to commit 20 hours per week towards this unit consistently throughout the trimester to achieve these outcomes.

Learning activities include attending two weekly seminars that guide your required learning for this unit, a classroom-based

studio workshop that will provide you opportunities to develop required skills and knowledge, and laboratory sessions for the equivalent of eight weeks across the trimester that provide you a practical means of learning control systems concepts. All sessions will be offered on-campus and online. Live sessions provide students with support and feedback for their learning. Attendance is highly recommended. Students are also expected to spend time reading or viewing the resources provided online and complete self-directed problem- based exercises and assessment tasks both individually and in teams.

All online students are required to attend interactive intensive week activities to complete their first laboratory sequence required for Summative Assessment Task 2 - Control systems lab portfolio and engage in team project related activities. For the compulsory intensive week, students are required to attend and participate in practical activities at the Waurn Ponds (Geelong) Campus for the scheduled sessions during the trimester intensive activities as detailed in the unit site. This intensive week activity is usually held in Week 6 of the trimester. Further details will be provided on the unit site.

Note

At Deakin, courses are delivered within a learning environment that provides all students with equitable and consistent access to facilities, infrastructure, resources, and support to assist student progress and achievement of learning outcomes.

We have introduced new terms to reflect learning activities to enhance your learning experience, aligning with our innovative **DeakinDesign learning principles and practices**. The new terms better reflect how teaching teams will guide you through your learning journey and the types of learning experiences you will have.

'Lectures' are the activities where teaching staff engage you through presentations with student participation.

In **'seminars'**, an educator will guide you in a smaller group of students through highly interactive discussions and activities.

Your units may also include '**practical experiences**' such as '**laboratory**', '**workshops**', '**clinical skills**' and more. These hands-on activities typically take place in specialised facilities with industry tools, equipment or technology to allow you to apply your knowledge practically.

Some other terms

If you see a '**meeting**' in your timetable, this is an optional drop-in session.

'Assessments' or **'team-based learning**' indicate an in-class evaluation of your skills or knowledge. A **'pre-assessment practice'** could be scheduled to prepare you for these assessments.

Find out more

Take a look at the **Learning activities webpage** for a full list of the terminology changes and reasons they were changed.
Unit learning resources

Your unit learning resources can be accessed from your unit site.
Prescribed text(s):

- R.C. Dorf and R.H. Bishop, 2021, Modern Control Systems Global 14th Edition, Mcgraw Hill.
- William Bolton, 2015, Programmable Logic Controllers, 6th Edition, Newnes, USA (e-copy freely available via the Deakin Library).

The texts and reading list for SEJ302 can be found via the University Library.

Note: Select the relevant trimester reading list. Please note that a future teaching period's reading list may not be available until a month prior to the start of that teaching period so you may wish to use the relevant trimester's prior year reading list as a guide only.
Essential learning resources

There are 2 prescribed texts for this unit as outlined above. Other essential resources will be provided on the unit site.

Recommended learning resources

The Deakin Software Library provides students with access to software that you may need or find useful for your study at Deakin.

The following recommended reading books are available on the Library's reading list. Students are not required to purchase these textbooks.

- Ogata, Modern Control Engineering, 5th edn, Prentice-Hall, Sydney, 2010**. (Recommended text)**
- Jack Golten and Andy Verwer, Control System Design and Simulation, McGraw-Hill, London. 1991. **(Recommended reading)**
- Howard Anton; Chris Rorres; Anton Kaul, 12th edition, Elementary linear algebra: Applications version, Wiley. **(Recommended reading)**

Where to access unit resources

Textbooks can be sourced from various outlets including direct from the publisher, online bookshops, or retailers. Limited copies of textbooks are also available on loan from the University Library.

Key dates for this study period

Trimester 2 teaching period begins	Monday 7 July 2025
Census Date	Friday 15 August 2025
Intra-trimester break	Monday 11 August - Sunday 17 August 2025
Trimester 2 teaching period ends	Friday 26 September 2025
Study period (end-of-unit assessment/examination preparation period)	Monday 29 September - Friday 3 October 2025
End-of-unit assessment/examination period	Monday 6 October - Friday 17 October 2025
Inter-trimester break (the period between trimesters)	Monday 20 October - Friday 31 October 2025
Unit results released	Friday 31 October 2025 (10.30 am)

Unit weekly activities

Week	Commencing	Topic	Special learning activities	Assessment due date
1	7 July 2025	Introduction to control systems, and systems modelling		
2	14 July 2025	Basis of Design, Reading P&ID's and systems modelling	Practical Labs	
3	21 July 2025	Systems modelling and introduction to system response	Practical Labs	**Assessment 1: Control systems theory test 1** due 27 July
4	28 July 2025	Giving and receiving interpersonal feedback, and system response	Practical Labs	**Assessment 2: Team control systems design project:** - Individual submission due 31 July - Feedback due 3 August
5^	4 August 2025	Reflective practice, and system performance specifications	Practical Labs	**Assessment 1: Control systems theory test 2** due 10 August

6	18 August 2025	Team formation, project planning, error and stability analysis	2-day Intensive On-Campus Practical Labs and Studios (online students only)	
7	25 August 2025	Socially responsible engineering industry case study, stability and intro to root locus design method	Practical Labs	**Assessment 3: Control systems lab portfolio** due 31 August
8	1 September 2025	Management milestone reviews, and automatic (PID) controller design	Practical Labs	**Assessment 1: Control systems theory test 3** due 7 September
9	8 September 2025	Work instructions, and automatic (PID) controller tuning	Practical Labs	**Assessment 2: Team control systems design project** - see unit site for booking details
10	15 September	Hardware selection, automatic controller design and validation	Practical Labs	**Assessment 1: Control systems theory test 4** due 14 September
11#	22 September 2025	Project Q&A and reflection, technical project support		
12	29 September 2025			**Assessment 2: Team control systems design project (team and individual)** due 5 October **Assessment 3: Control systems lab portfolio** due 29 September Study period
13-14	6 October 2025			End-of-unit assessment and exam period

^ Intra-trimester break: **Monday 11 August – Sunday 17 August 2025**
AFL Grand Final Eve public holiday: **Friday 26 September 2025** - University closed